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CAREER SUMMARY

Currently pursuing a Master's in Artificial Intelligence and Machine Learning at the University of Adelaide, with strong proficiency in Python, C++, C, and Ruby. Experienced in OpenCV, CNNs, NLP, LLMs, speech recognition, TTS, and audio signal processing. Skilled in deploying machine learning solutions on AWS and other cloud platforms. Passionate about building scalable ML pipelines and integrating advanced algorithms into practical applications across computer vision, natural language processing, and audio domains. Open to roles in AI research and development.

EDUCATION

2024 - 2026

University of Adelaide

Masters in Artificial Intelligence and Machine Learning
GPA: 6.2

2020 - 2024

Chandigarh University

B.E. in Computer Science and Engineering (AI/Machine Learning)
GPA: 8.39

2019 - 2020

CBSE Board (NP Co-ed, Lodhi Estate, New Delhi-110002)

12th (Senior Secondary)
Percentage: 81.5%

EXPERIENCE

June 2022 -
August 2022

Backend Web Development Intern

Curve Tomorrow

- Built and maintained scalable backend features in Ruby on Rails for internal tools and admin-facing web applications.
- Created a user activity logging system to improve auditing and simplify backend issue tracing.
- Developed a dynamic Content Management System (CMS) enabled non-technical users to edit and publish content, reducing turnaround time from days to minutes.
- Collaborated with distributed teams in agile sprints to design RESTful APIs, manage relational databases, and optimize server-side logic.
- Implemented on projects user authentication, CRUD functionality, blog engines, and MVC architecture, applying best practices in Rails development across healthcare domain.

PROJECTS

GPT-2 From Scratch — (Live: [Link](#), GitHub: [Link](#), HuggingFace: [Link](#))

Large Language Model Training

- Pretrained GPT-2 Small (124M) from scratch on Ultra-FineWeb Edu (5B tokens) using Kaggle T4 GPUs, implementing an end-to-end LLM training pipeline.
- Built an optimized training stack with Flash Attention, FP16 AMP, AdamW (GPT-3 style), cosine LR decay, gradient accumulation, DDP, and robust checkpointing.
- Achieved 28.34% accuracy on HellaSwag, matching the original GPT-2 Small baseline and validating training correctness.
- Performed Supervised Fine-Tuning (SFT) on a custom instruction dataset; experimented with DPO/RLHF, analyzing overfitting limits in small models.
- Deployed the model as a Gradio web app for real-time public inference.

Rag Implementation ([Link](#))

Using Lang Chain

- **RAG (Retrieval Augmentation Generation):** Extract information from PDFs, Docs, or websites effortlessly with Apsara's RAG system.
- Integrated support for both local embeddings and GPT-based embeddings powered Ollama and Open AI for enhanced language understanding.

Face Mask Detection ([Link](#))

Using Transfer Learning and Computer Vision

- Detects whether a person is wearing a mask or not (with multi-faces in image enabled as well), trained using TensorFlow & MobileNetv2 architecture.
- Deployed an email notification system to alert administrators when mask enforcement is violated or a mask is not worn for over 30 seconds.

SKILLS

- **Technical Skills:** Computer Vision, Data Structures and Algorithms, Web Scraping, Natural Language Processing (NLP), Prompt Engineering, Database Management System, Big Data, Command-Line Interface, Web Development (HTML, CSS), Cloud Technologies, Unix, AWS, Data Analysis, Machine Learning.
- **Tools and Technologies:** OpenCV, Selenium, Ruby on Rails, Py-Torch, Flask, Pandas, NumPy, Git, Lang Chain, MySQL, TensorFlow.
- **Programming Languages:** Python, C++, C, Java, JavaScript, Ruby.
- **Soft Skills:** Communication, Teamwork & Collaboration, Leadership, Decision-Making, Self-Motivation, Work Ethic, Stress Management, Time Management.

CERTIFICATES

- [Machine Learning](#)
- [Tools for Data Science](#)
- [Supervised Machine Learning: Regression and Classification](#)
- [Advanced Learning Algorithms](#)
- [Databases and SQL for Data Science with Python](#)
- [Crash Course on Python](#)
- [Using Python to Interact with the Operating System](#)
- [Linux Fundamentals](#)